Lab: Renormalization Group and Scale-Free Active Inference

> **Learning Goal:** Apply RG concepts to Active Inference at multiple scales, analyzing coarse-graining, criticality, and multi-scale organization.

Part 1: Multi-Scale Markov Blanket Identification

Exercise: For the human body, identify the Markov blanket at each organizational scale:

Scale	System	External States	Sensory States	Active States	Internal States
Molecular	Single protein	Solvent molecules, ligands	Binding sites, allosteric sites	Conformational changes, enzymatic activity	Intramolecular dynamics
Cellular	Neuron	Extracellular environment	Receptors, ion channels, synaptic inputs	Axon terminals, neurotransmitter release	Intracellular signaling, gene expression
Circuit	Cortical column	Other columns, subcortical input	Input layer neurons (L4)	Output layer neurons (L5/6)	Internal processing layers (L2/3)
Organ	Brain	Body, environment	Sensory nerves	Motor nerves, endocrine	Cortical and subcortical processing
Organism	Person	Physical/ social environment	Sense organs (eyes, ears)	Muscles, voice, hands	Brain, viscera
Social	Family	Other families, institutions	Communication received	Communication sent, actions	Shared beliefs, family dynamics

Write a 200-word analysis: At which scale does the Markov blanket identification become most controversial? Why?

Write your response here

Part 2: Coarse-Graining Exercise

Learning Goal: Perform a conceptual RG transformation on a neural system.

Exercise: Consider a network of 100 spiking neurons. Perform two levels of coarse-graining:

Level 0 (Microscopic): 100 individual neurons with spiking dynamics

- States: individual membrane potentials $V_1, V_2, ..., V_{100}$
- Interactions: synaptic weights w_{ij}

Level 1 (Mesoscopic): Group into 10 neural populations of 10 neurons each

- States: mean firing rates $r_1, r_2, ..., r_{10}$
- Effective interactions: population coupling J_{IJ} = some function of {w_{ij}}

Level 2 (Macroscopic): Group into 2 brain regions of 5 populations each

- States: regional activity levels R_1, R_2
- Effective interaction: inter-regional coupling

At each level: (a) What information is lost? (b) What information is preserved? (c) Does the generative model structure (A, B matrices) change form?

Write your response here

Part 3: Criticality Analysis

Learning Goal: Evaluate the evidence for neural criticality.

Exercise: Evaluate the following evidence for brain criticality:

Evidence	Finding	Supports Criticality?	Alternative Explanation
Neural avalanches	Power-law size distribution with exponent ~3/2	Yes — matches critical Ising model	Could be subcritical with weak correlations
1/f noise	Power spectrum ~1/f across broad frequency range	Yes — scale invariance	Could result from multiple uncorrelated processes
Long-range correlations	fMRI shows correlations across distant brain regions	Yes — divergent correlation length	Could reflect anatomical connectivity
Maximal dynamic range	Neural systems at criticality respond to widest range of input intensities	Yes — optimal sensitivity	Could be selected for without criticality
Information capacity	Criticality maximizes mutual information between input and output	Yes — optimal coding	Other mechanisms achieve good coding

Write a 300-word assessment: Is the brain at criticality, near criticality, or is this a misleading framework?

Write your response here

Part 4: Society as Active Inference

> **Learning Goal:** Apply scale-free Active Inference to social systems.

Exercise: Model a political election as Active Inference at the social scale:

1. **Social generative model**: What is the "shared model" of a political party? (Beliefs about the economy, society, threats, opportunities)
2. **Social prediction errors**: What events generate prediction errors in the social model? (Unexpected economic data, scandals, crises)
3. **Social active inference**: How does the party act to minimize free energy? (Policy proposals = active states; media = sensory management)
4. **Social learning**: How does the party's model update after an election loss?
5. **Echo chambers as precision traps**: How do echo chambers function as excessive precision on prior beliefs, preventing model updating?

Write your response here

Part 5: Reflection

In 300 words, reflect: The scale-free claim of Active Inference — that the same principles apply from molecules to societies — is either the framework's greatest strength or its greatest weakness. If it applies to everything, does it predict anything? Compare this to other "universal" frameworks (thermodynamics, natural selection, information theory). Are they similarly general, and are they similarly useful?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Multi-scale identification	Nested Markov blankets
2	RG transformation	Coarse-graining neural systems
3	Evidence evaluation	Neural criticality
4	Social application	Scale-free framework
5	Philosophical critique	Universality vs. predictive power